



■ CS177 Bioinformatics: Software Development and Applications

Lecture 15: AI for Protein Design

Jie Zheng (郑杰)

PhD, Associate Professor

School of Information Science and Technology (SIST), ShanghaiTech University

Spring, 2025



Learning objectives



- At the end of this lecture, you would be able to:
 - Define the objectives and problems of protein design
 - List and explain a few in silico metrics for the evaluation of designed proteins
 - Describe a few AI models for protein design based on protein language models (ProGen, ESM3)
 - Explain the ideas and steps of RFdiffusion and ProteinMPNN
 - Describe how to guide (or steer, or prompt) AI models to generate protein structures or sequences with desirable properties





Overview of AI for Protein Design

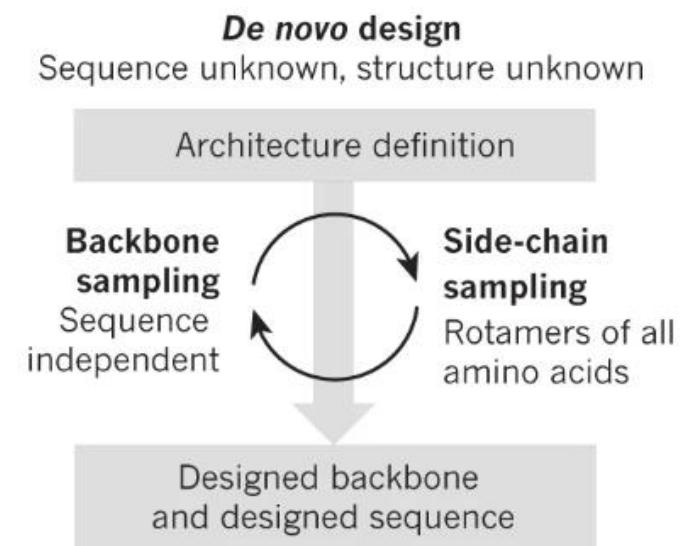
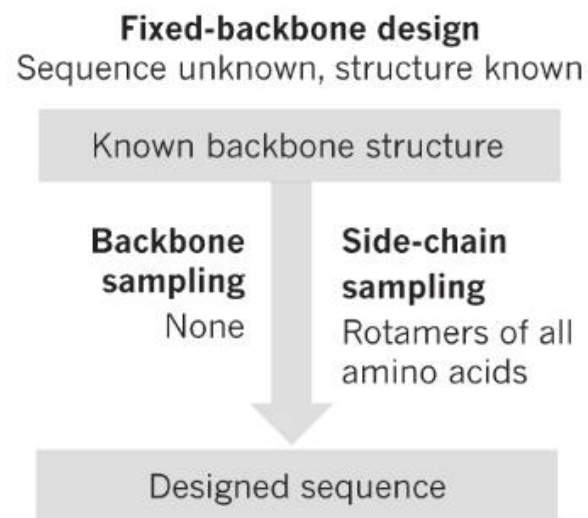
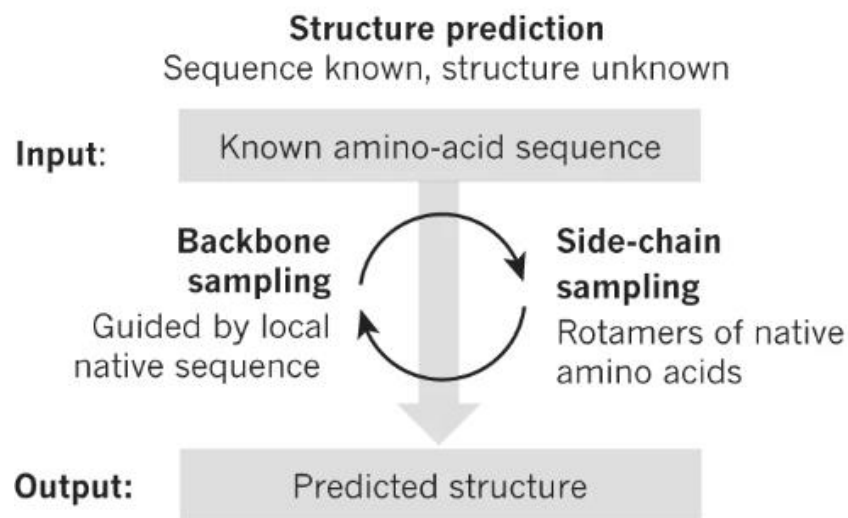
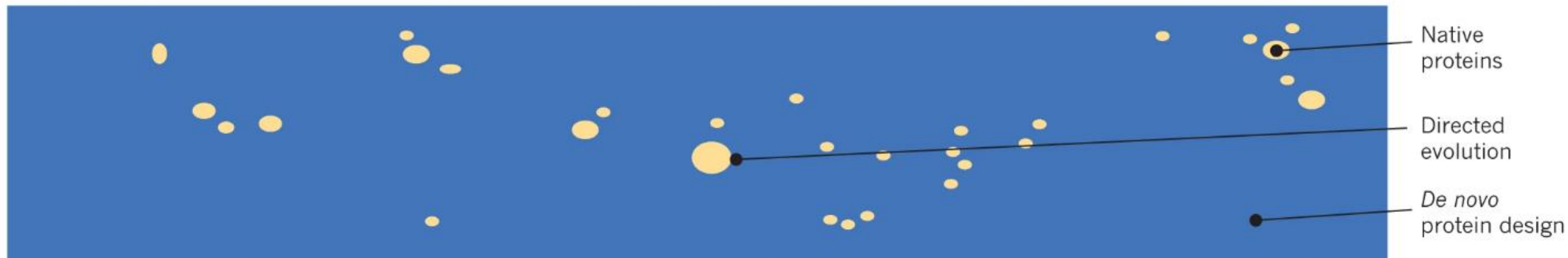




Protein Design



上海科技大学
ShanghaiTech University



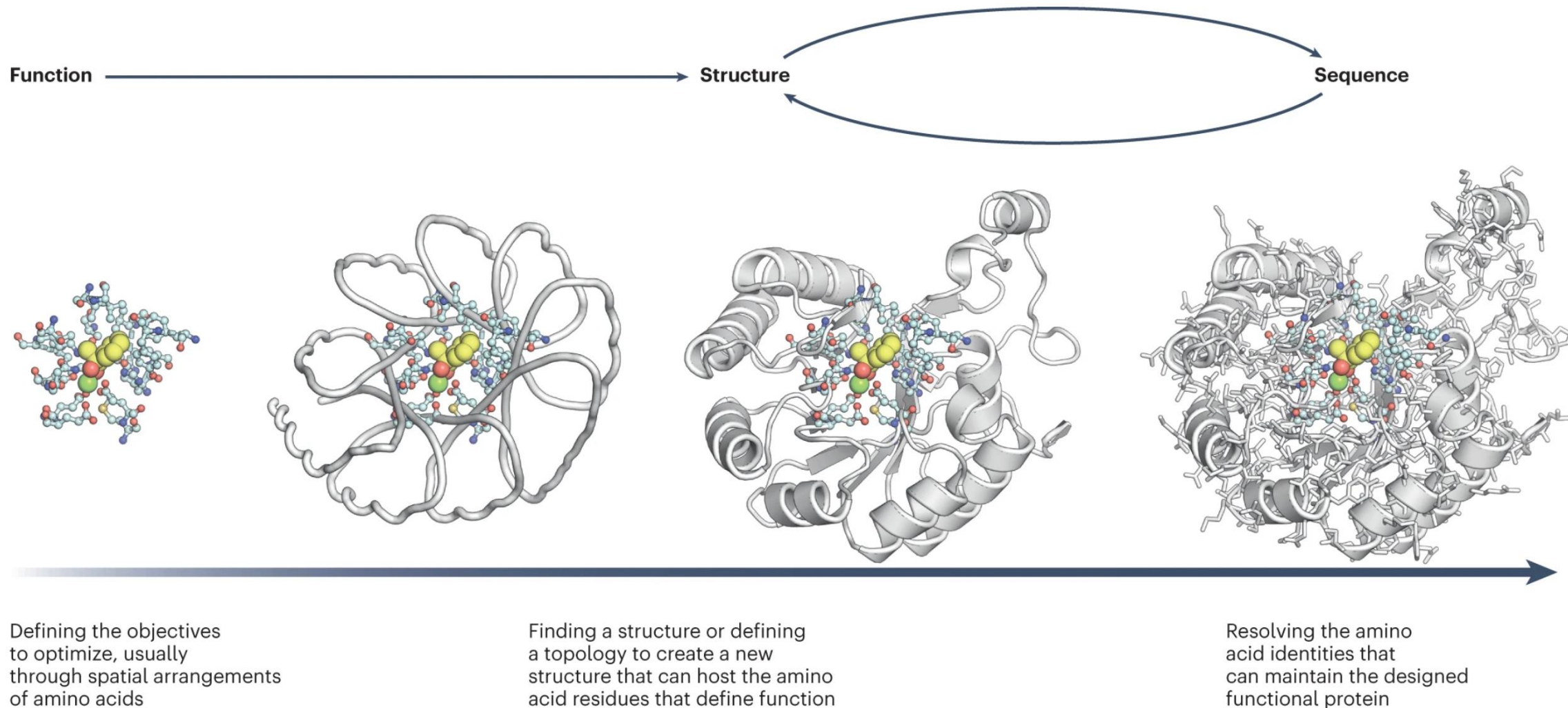
P.-S. Huang, S. E. Boyken, D. Baker. "The coming of age of *de novo* protein design" , Nature 2016.

立志成才 报 国 裕 民

The central dogma of de novo protein design

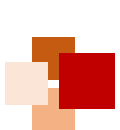


上海科技大学
ShanghaiTech University



Chu, A.E., Lu, T. & Huang, P.S. Sparks of function by de novo protein design. *Nature Biotech* **42**, 203–215 (2024).

立志成才 报 国 裕 民

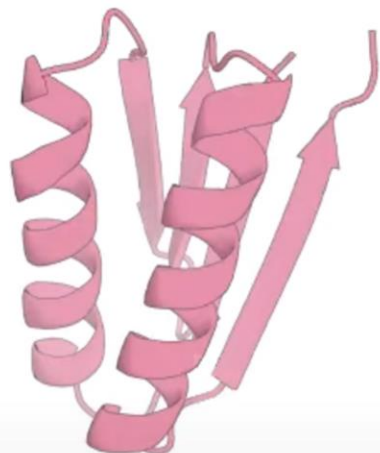


General protein design process

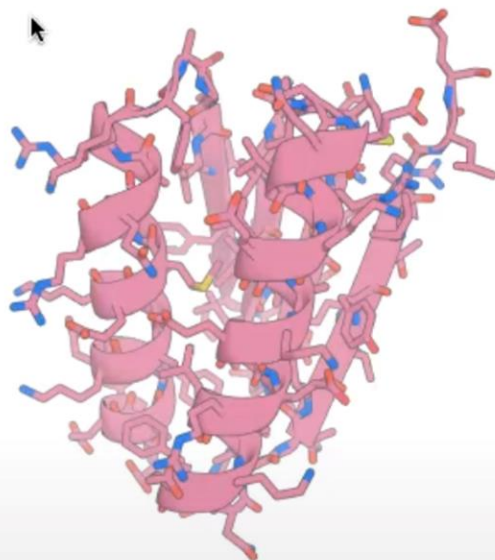


上海科技大学
ShanghaiTech University

Backbone
Generation

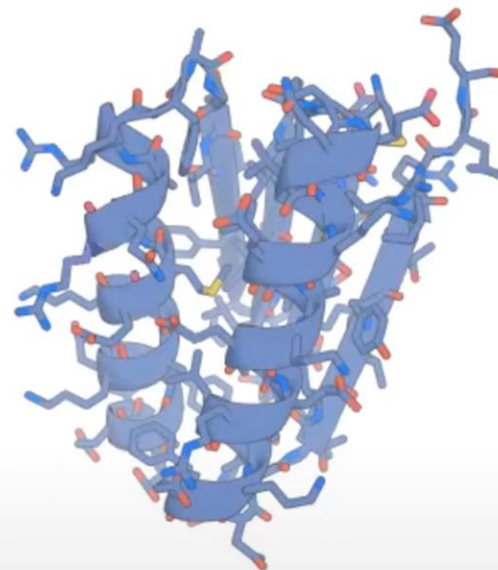


Sequence
Design



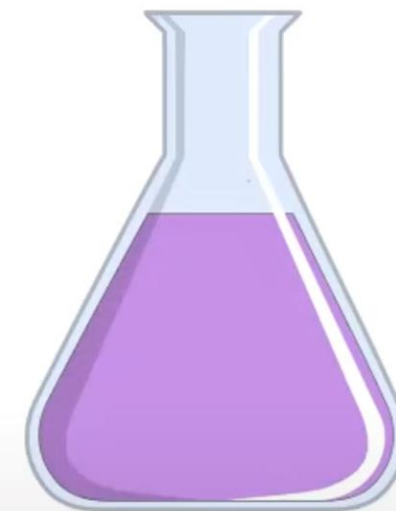
ProteinMPNN

Computational
Filtering



AlphaFold2
RoseTTAFold
ESMFold
OmegaFold

Experimental
Characterisation

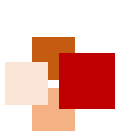


*96-384 well
purification
(~ 1 week)*

<https://www.youtube.com/watch?v=wIHwHDt2NoI>



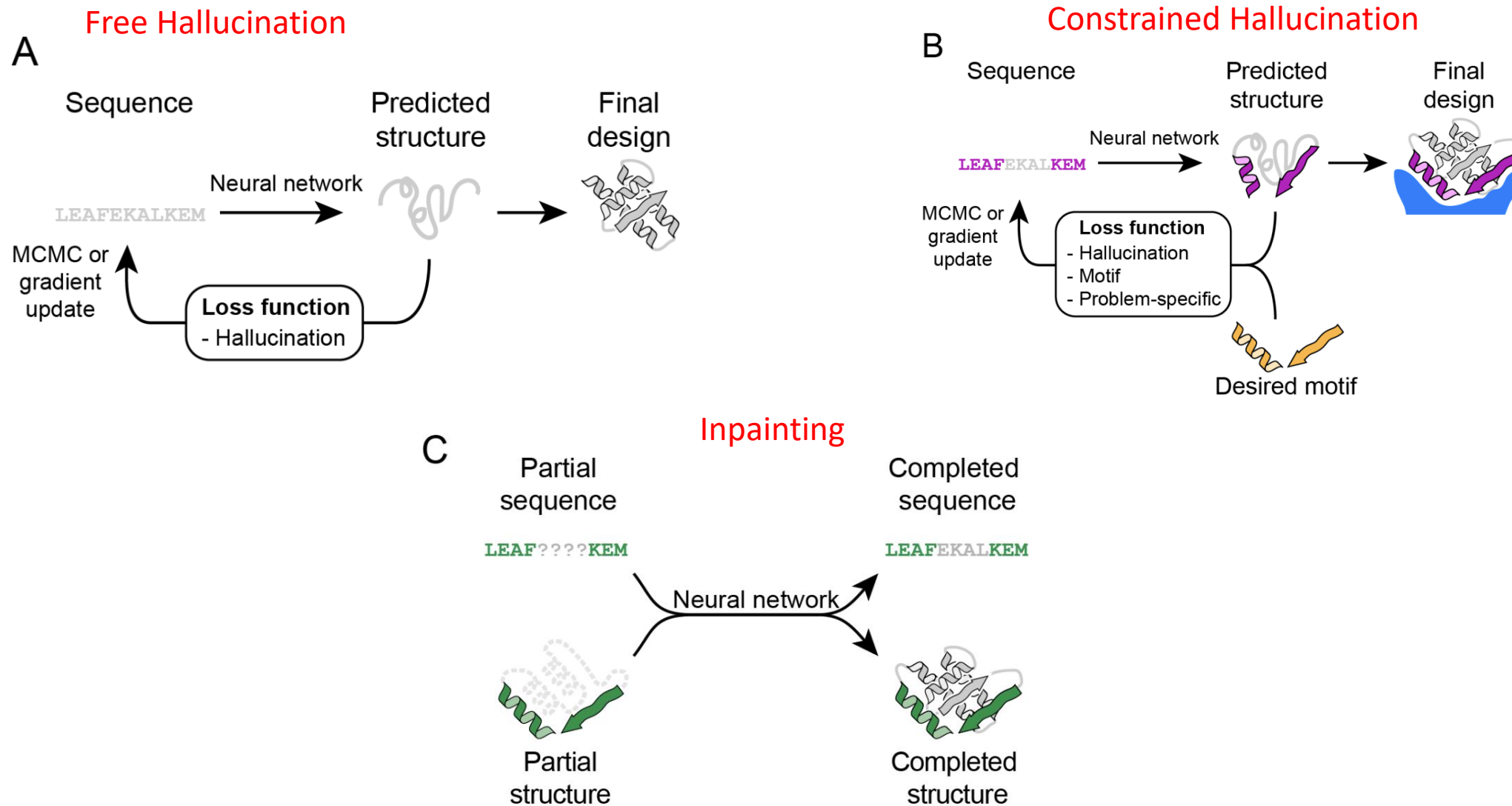
立志成才 报國裕民



Methods for protein function design



上海科技大学
ShanghaiTech University



J. Wang, S. Lisanza, D. Juergens, D. Tischer, et al. bioRxiv 2021.

立志成才 报 国 裕 民



Evaluation



- Some *in silico* metrics:
 - **Designability**: The total number of amino acid sequences that can fold into a target protein structure.
 - **Self-consistency**: Sequences for the protein are generated with a sequence design model (e.g. [ProteinMPNN](#)). These sequences are then fed into a structure prediction model (e.g. [AlphaFold 2](#)) and the [agreement](#) between the predicted structure and the original protein structure is measured.
 - **Novelty**: It measures the extent to which the generated samples differ significantly from the reference set, e.g. the maximum [TM-score](#) between the generated and the training set of proteins or between the generated set and any chain in the Protein Data Bank (PDB) using [FoldSeek](#).
 - **Coverage**: It measures the ability of the model to sample the full topological space of proteins.
 - **Diversity**: The internal [variety](#) of generated proteins and is measured to ensure that generative models cover all relevant modes of the target distribution and avoid phenomena such as 'mode collapse'.
 - **Efficiency**: the ability to achieve the desired protein structure or function with [minimal computational resources](#), e.g. the time and memory usage for model inference.
- Experimental validation:
 - X-ray Crystallography
 - Cryo-electron microscopy (Cryo-EM)
 - Functional studies

From Joshua Southern et al. "Evaluation metrics for protein structure generation", ICML Workshop on Deployable Generative AI, 2023.





Protein Language Model-based Methods

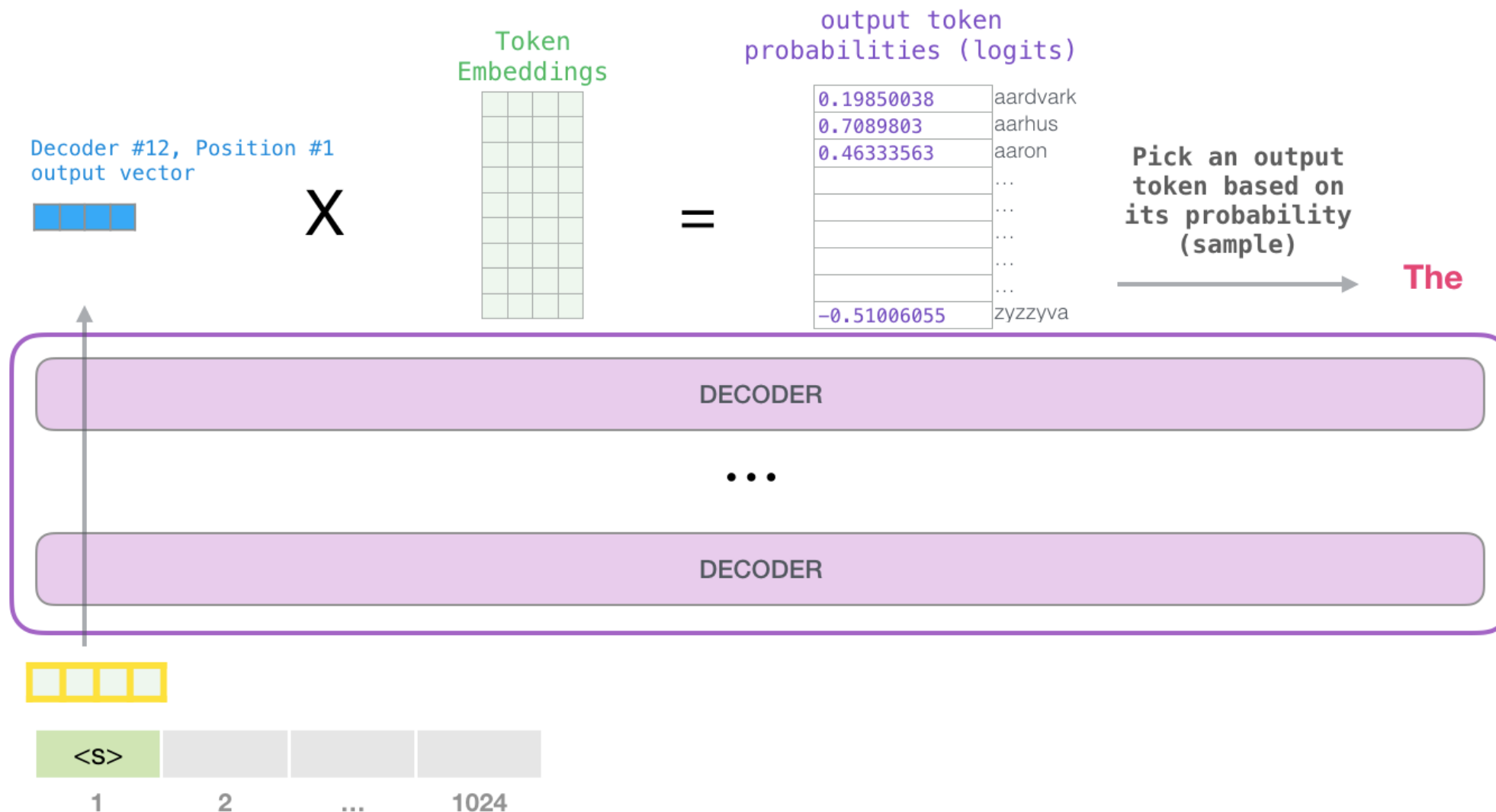




Recap: Language Models (LMs)

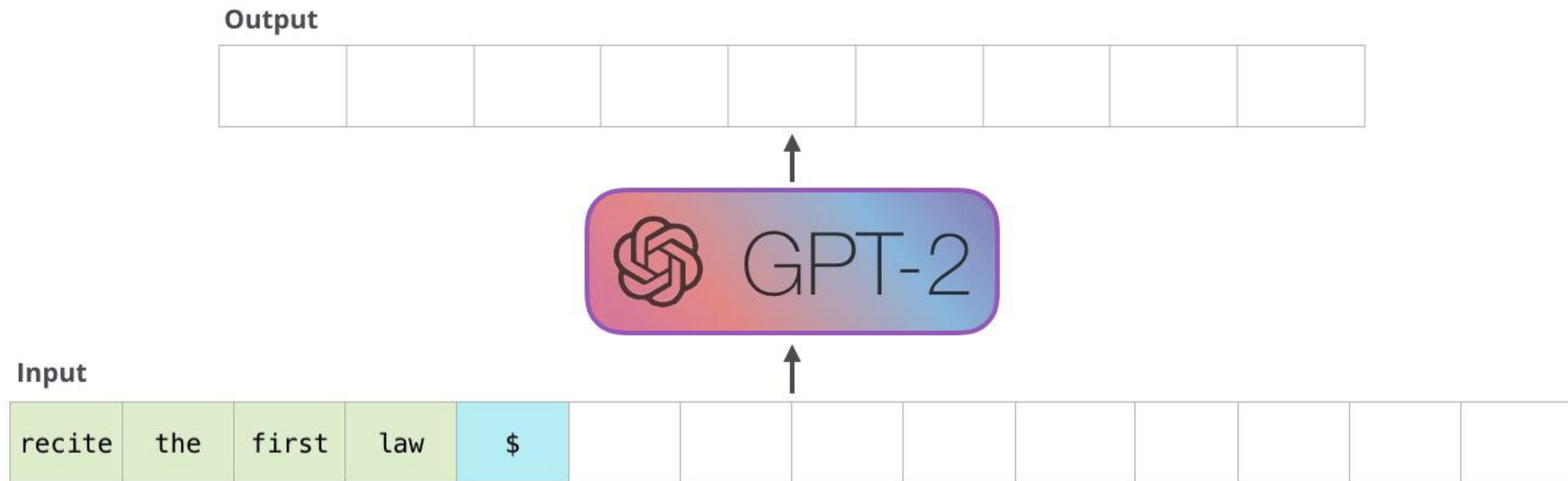


An LM aims to predict the probabilities of future (or missing) tokens



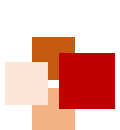


Auto-regressive language models



GPT-2 generates sequences by “auto-regression”





CTRL (Conditional Transformer Language)



上海科技大学
ShanghaiTech University

Language modeling: Given example sequences of the form $x = (x_1, \dots, x_n)$, the goal is to learn $p(x)$, i.e. the probability distribution of the sequences, decomposed using the **chain rule of probability**:

$$p(x) = \prod_{i=1}^n p(x_i | x_{<i})$$

Learning objective: To train a neural network with parameters θ to minimize the negative log-likelihood over a dataset D :

$$\mathcal{L}(D) = - \sum_{k=1}^{|D|} \log p_{\theta}(x_i^k | x_{<i}^k)$$

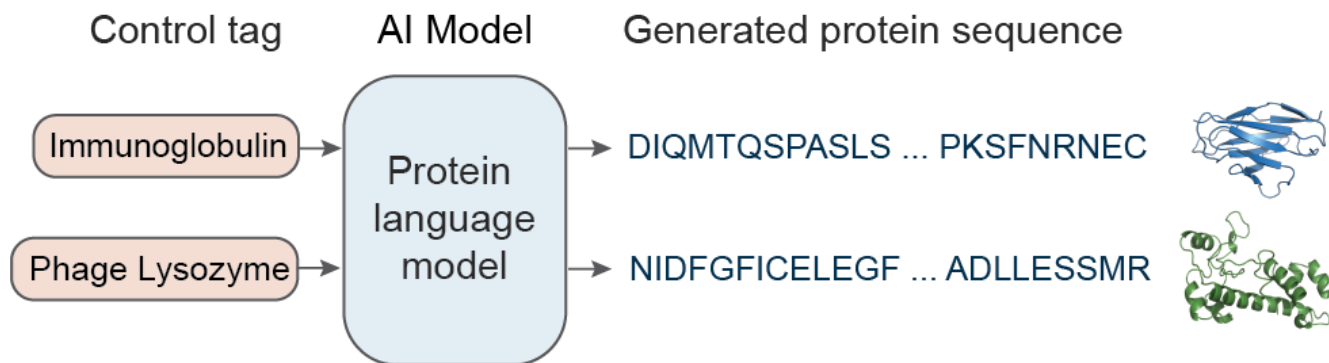
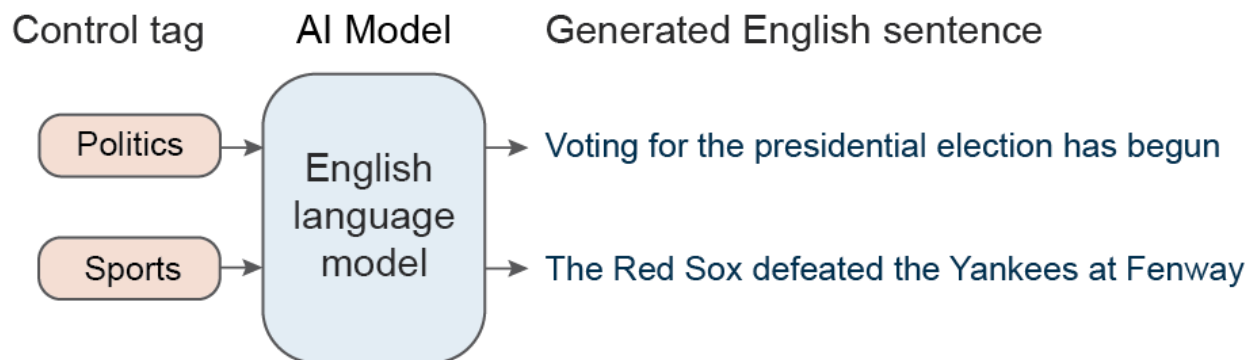
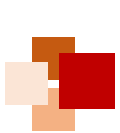
CTRL is a Transformer-based conditional language model that is conditioned on a **control code c** and learns the probability distribution $p(x|c)$

$$p(x|c) = \prod_{i=1}^n p(x_i | x_{<i}, c) \quad \mathcal{L}(D) = - \sum_{k=1}^{|D|} \log p_{\theta}(x_i^k | x_{<i}^k, c^k)$$

Keskar, N. S., McCann, B., Varshney, L. R., Xiong, C., and Socher, R. CTRL: A conditional transformer language model for controllable generation. arXiv preprint arXiv:1909.05858, 2019.

立志成才 报國裕民





$$\mathcal{L}(D) = - \sum_{k=1}^{|D|} \log p_{\theta}(x_i^k | x_{<i}^k)$$

Training: Negative log likelihood minimization

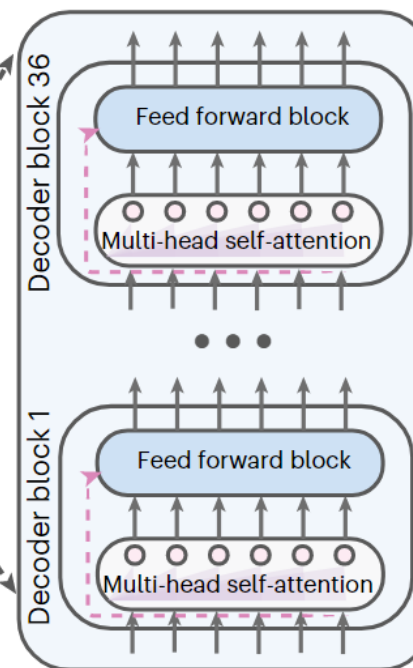
Next amino acid prediction

$a_1 \ a_2 \ a_3 \ a_4 \ \dots$

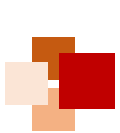
Transformer decoder

$c \ a_1 \ a_2 \ a_3 \ \dots$

Control tag Amino acids



Ali Madani et al, Nat Biotechnol 2023



Example of training and fine-tuning data



Sample sequences with associated control tags provided during the training and fine-tuning

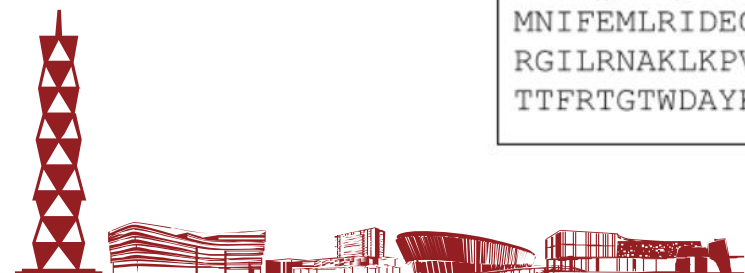
Template: $\langle c_1 \rangle \langle c_2 \rangle \dots \langle c_N \rangle a_1 a_2 a_3 a_4 a_5 \dots$

Training Sample

```
<Metazoa><Chordata><Mammalia><Rodentia><Muridae><Rattus><Rattus><Norvegicus>  
<NAD>  
<Translocase>  
<Iron><Iron-sulfur><2Fe-2S>  
<Mitochondrion><Mitochondrion inner membrane>  
<Transport><Electron transport><Respiratory chain>  
MFSLALRARASGLTAQWGRHARNLHKTAVQNGAGGALFVHRDTPENNPDTPFDFTPENYERIEAIVRNYPEGHRAA  
AVLPVLDLAQRQNGWLPISAMNKVAEVLQVPPMRVYEIVATFYTMYNRKPVGKYHIQVCTTTPCMLRDSISILETLQ  
RKLGIKVGETTPDKLFTLIEVECLGACVNAPMVQINDDYIEDLTPKDIEEIIDELRAGKVPKPGPRSGRFCCEPAG  
GLTSLTEPPKGPFGVQAGL
```

Fine-tuning Sample

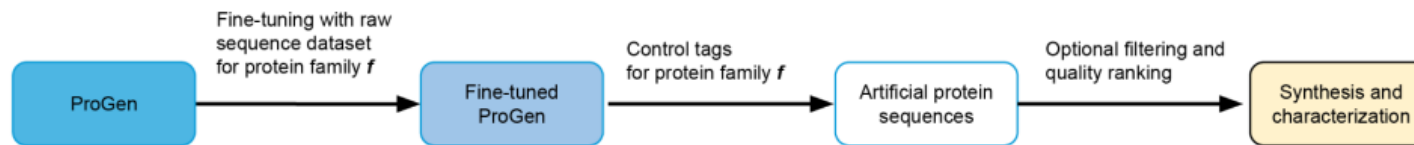
```
<Phage Lysozyme>  
MNIFEMLRIDEGLRLKIYKDTEGYTIGIGHLLTKSPSLNAAKSELDKAIGRNCNGVITKDEAEKLFNQDVDAAV  
RGILRNAKLKPVYDSLDAVRRCALINMVFQMGETGVAGFTNSLRMLQQKRWDEAAVNLAWSRWYNQTPNRAKRVI  
TTFRTGTWDAYKNL
```



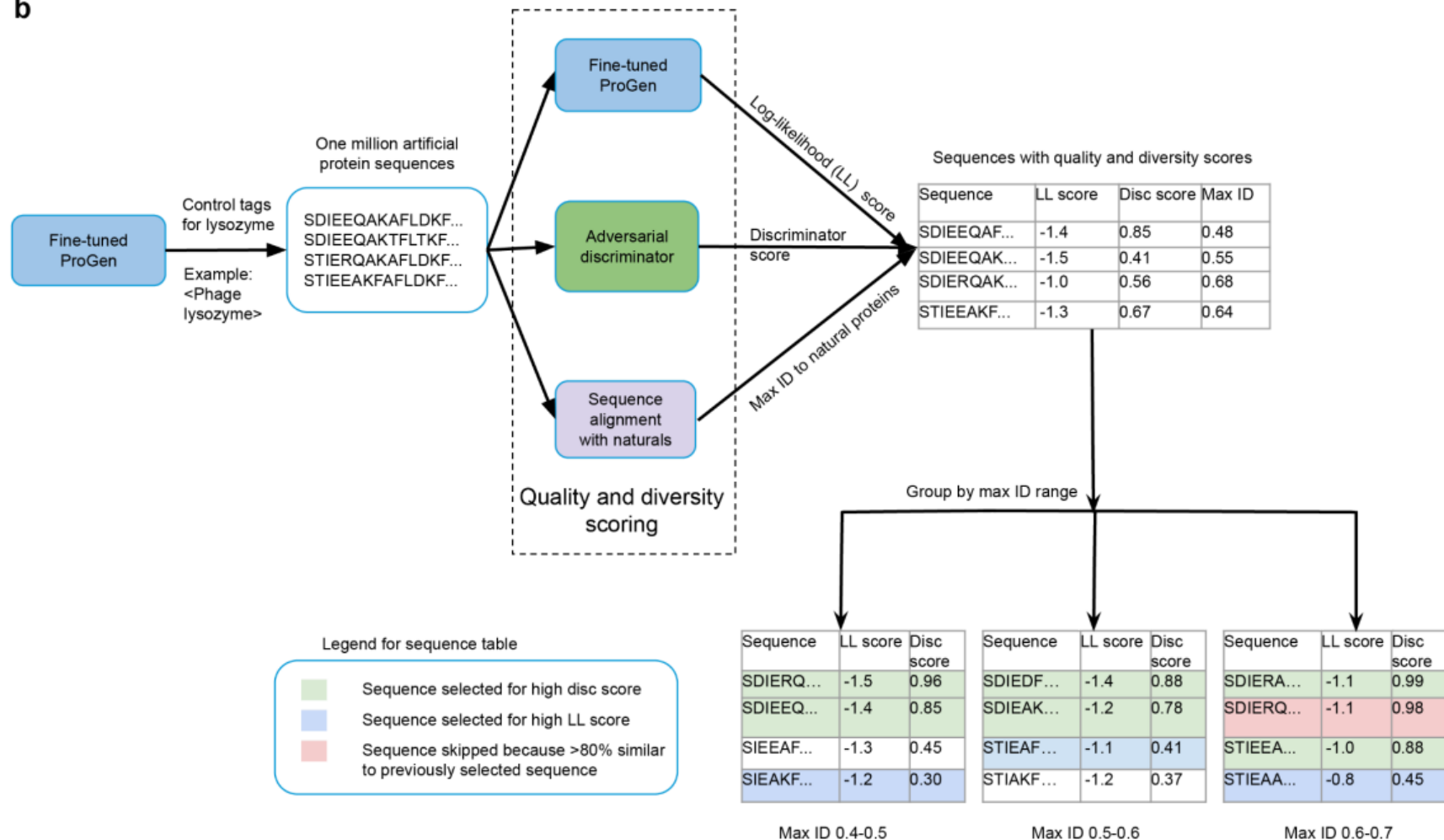
Generating and selecting proteins for synthesis



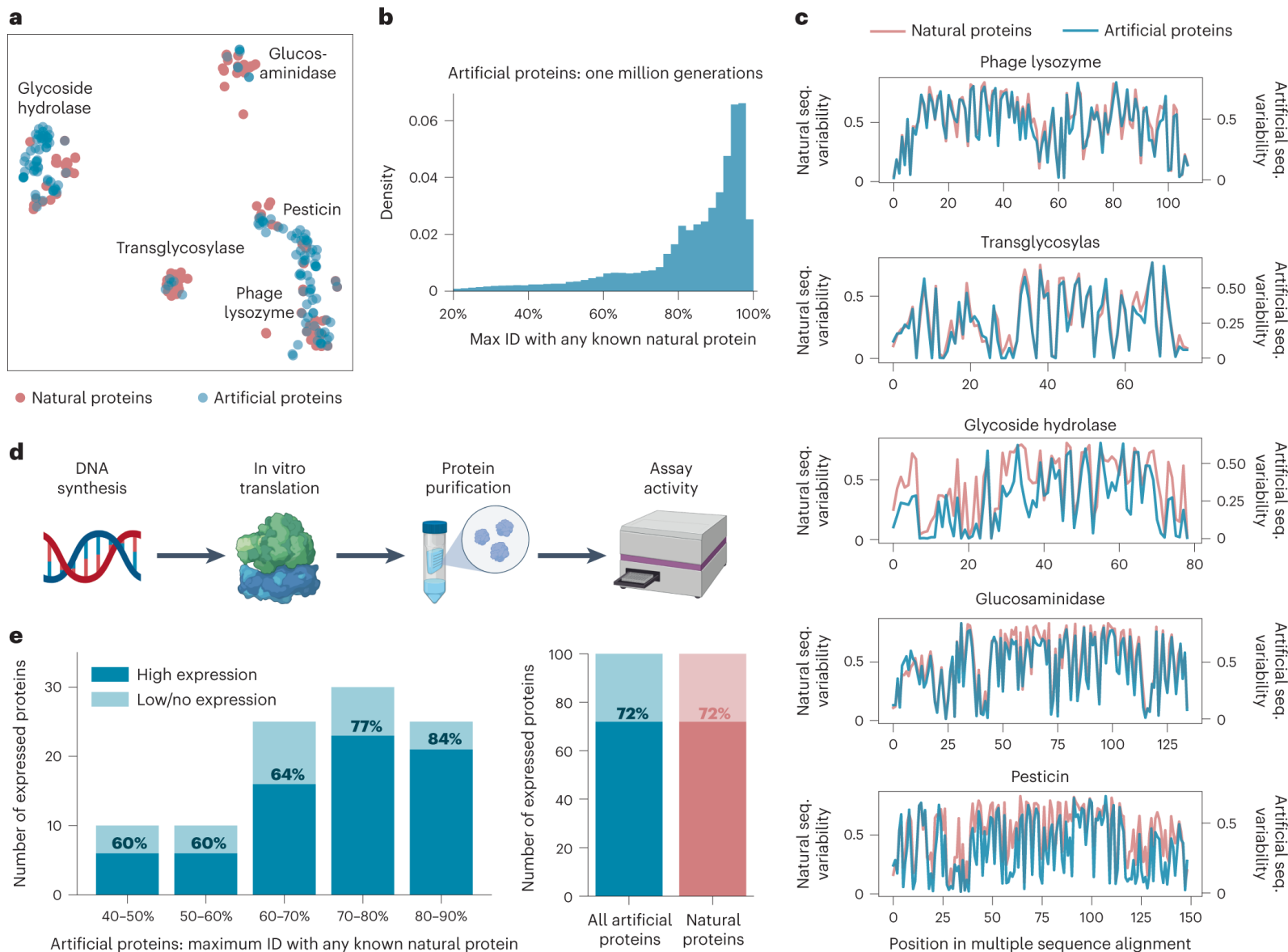
a



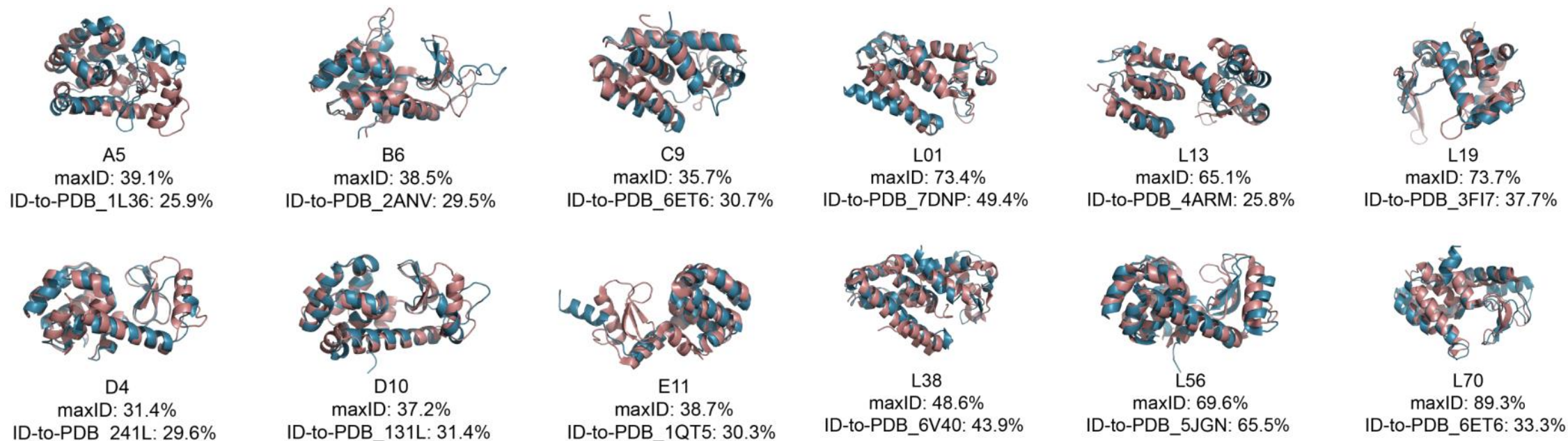
b



Generated proteins are diverse and express well



Artificial proteins are similar to natural proteins in structures



■ Natural protein

■ Artificial protein

Ali Madani et al, Nat Biotechnol 2023



Article

ProGen2: Exploring the boundaries of protein language models

Erik Nijkamp,^{1,5} Jeffrey A. Ruffolo,^{2,3,5} Eli N. Weinstein,⁴ Nikhil Naik,¹ and Ali Madani^{1,3,6,*}

¹Salesforce Research, Palo Alto, CA, USA

²Program in Molecular Biophysics, The Johns Hopkins University, Baltimore, MD, USA

³Profluent Bio, Berkeley, CA, USA

⁴Data Science Institute, Columbia University, New York, NY, USA

⁵These authors contributed equally

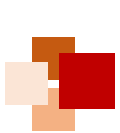
⁶Lead contact

*Correspondence: ali@profluent.bio

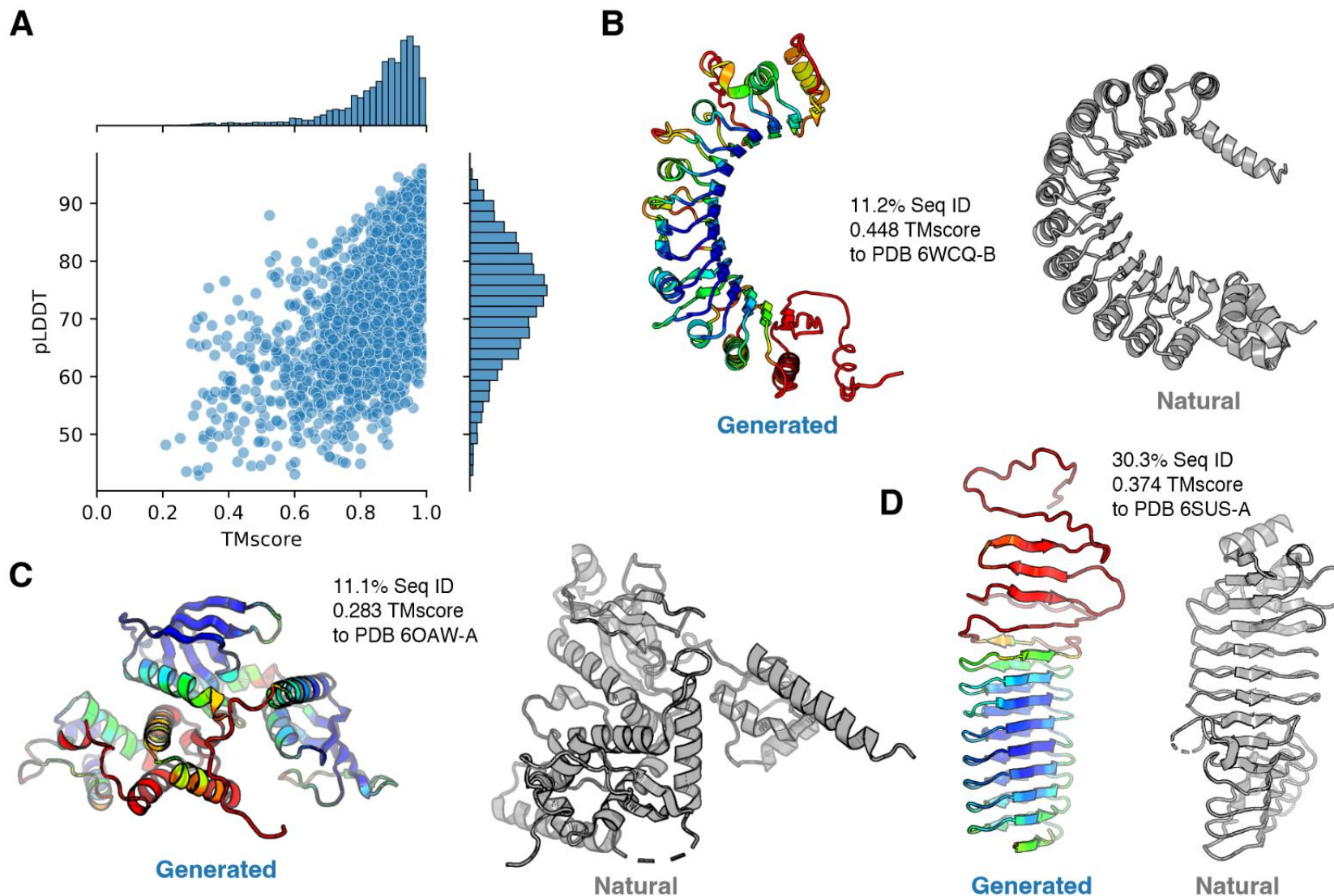
<https://doi.org/10.1016/j.cels.2023.10.002>

Nijkamp E, et al. ProGen2: Exploring the boundaries of protein language models. Cell Systems, 14(11):968-978.e3, 2023





Generation from pre-trained models



Weaknesses of ProGen



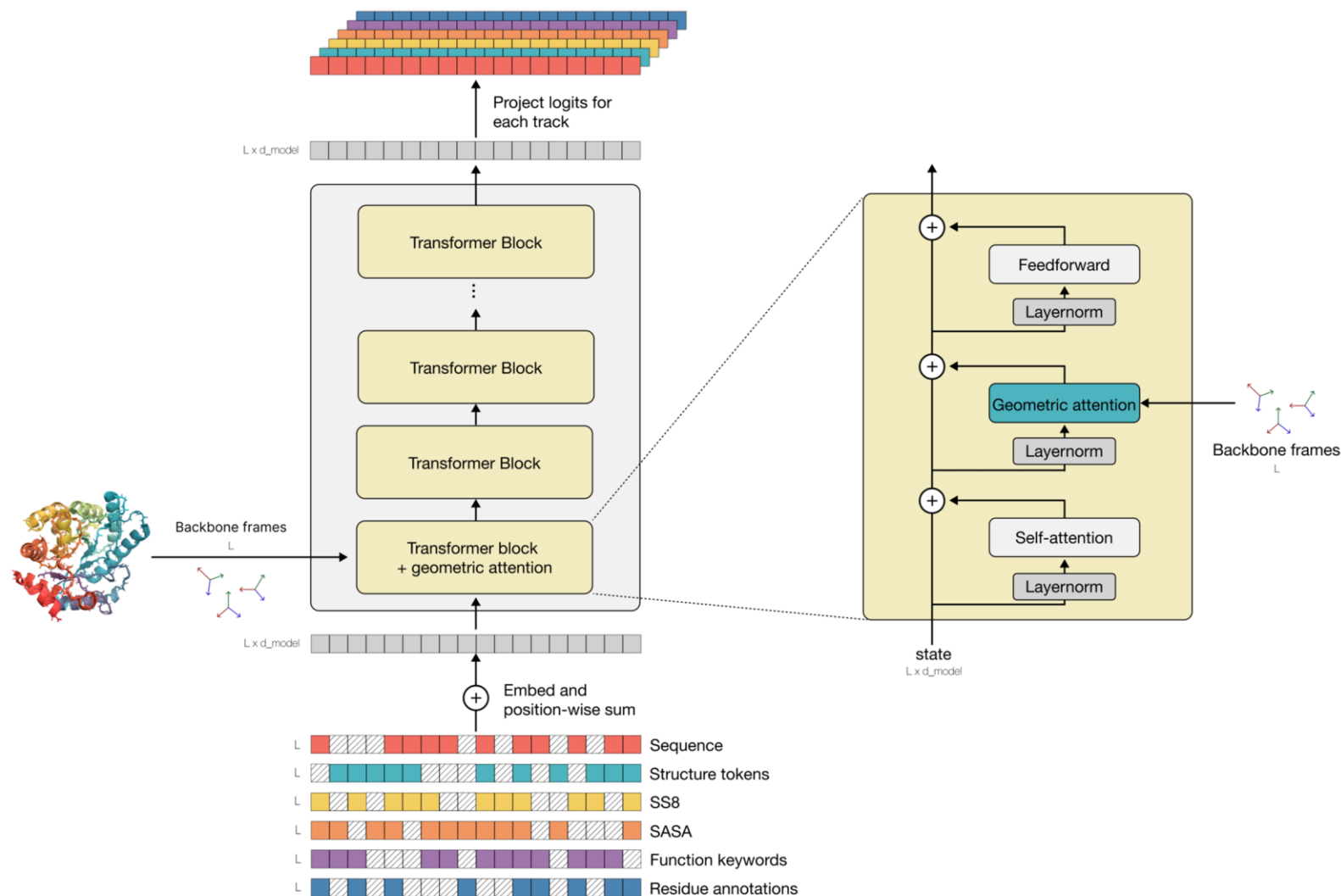
- **ProGen1:**

- The effect of model for a specific protein family is **dependent on** the quantity and quality of the specific fine-tuning **dataset**
- It is **time-consuming** to add control tags to sequences of the pretrained dataset

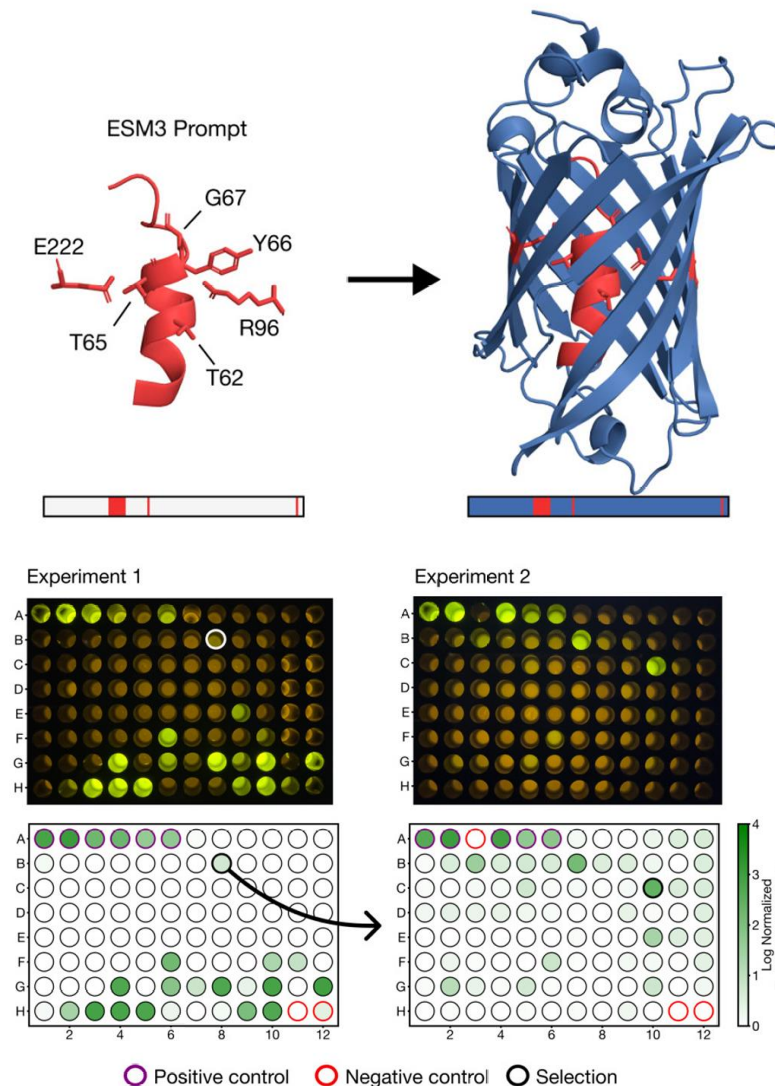
- **ProGen2:**

- ProGen2 cannot generate sequences containing **specific** structural and functional domains without fine-tuning
- Compared with current top Large Language Models (LLMs), the model size and dataset size of ProGen2 are still **not large enough**





ESM3 generates a new fluorescent protein with a chain of thought



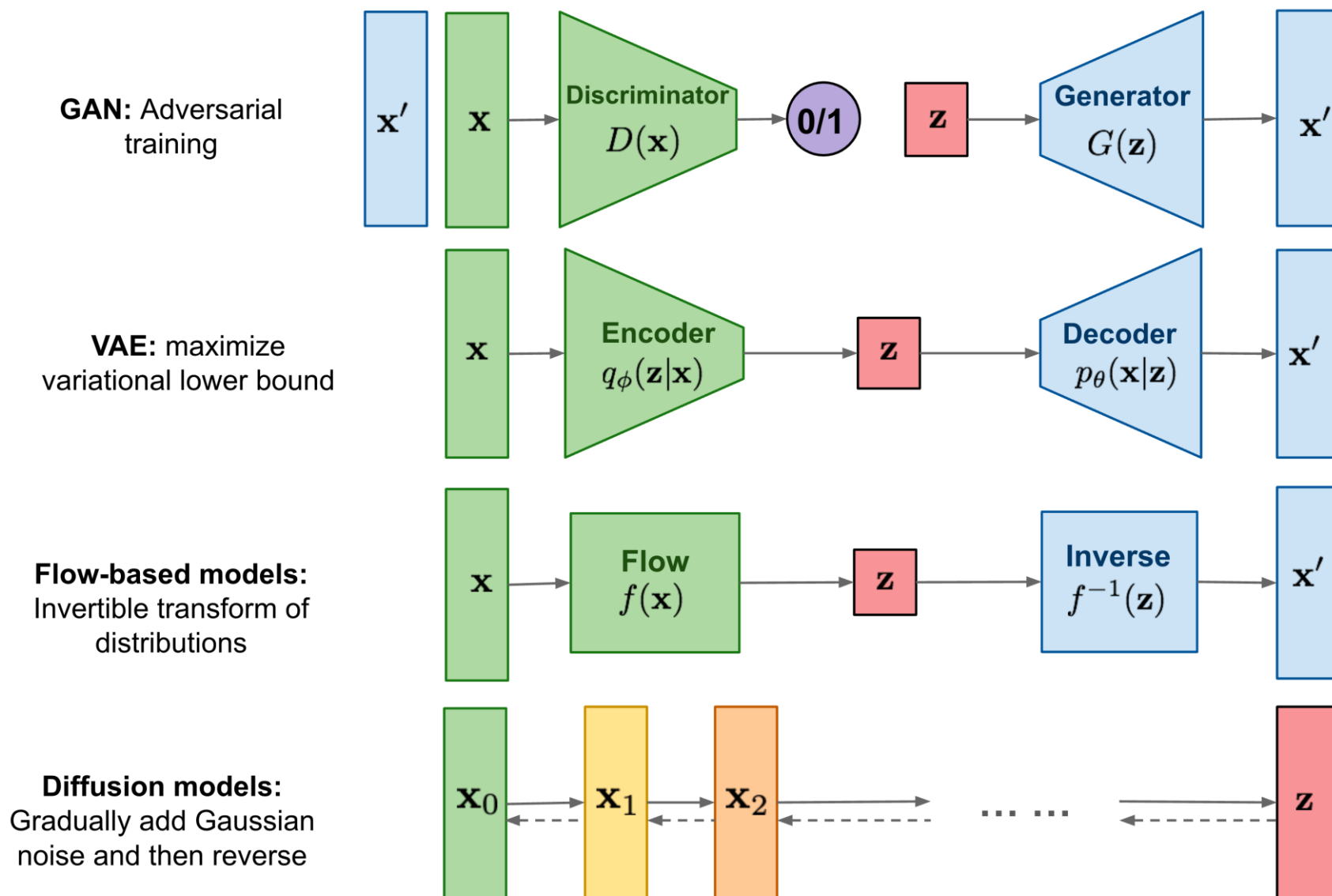
Thomas Hayes, et al. Simulating 500 million years of evolution with a language model, bioRxiv 2024.

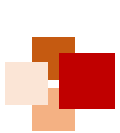


Diffusion Model-based Methods



Generative models

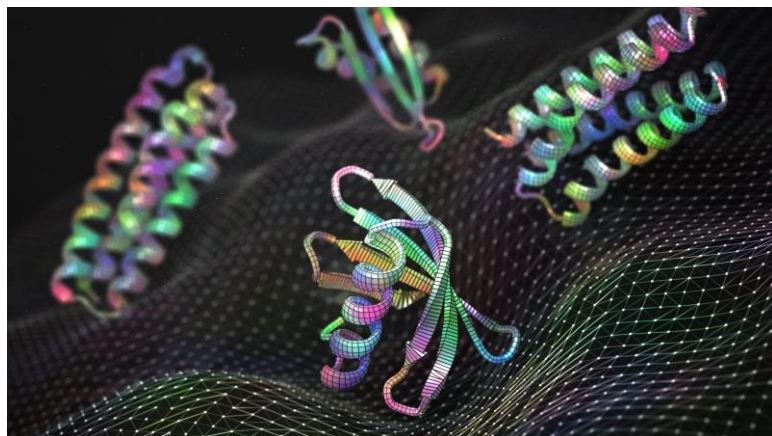




Diffusion in computational biology



后AlphaFold2时代，研究者们努力发掘更大的AF2落地场景，即**从结构预测走向结构设计/生成**



- 其中，Diffusion正是一个良好的赋能手段。2022-2023年也是Diffusion在计算生物学领域开始爆发的两年
- 主要集中于：**小分子生成、分子对接、蛋白质生成/蛋白质序列设计**等

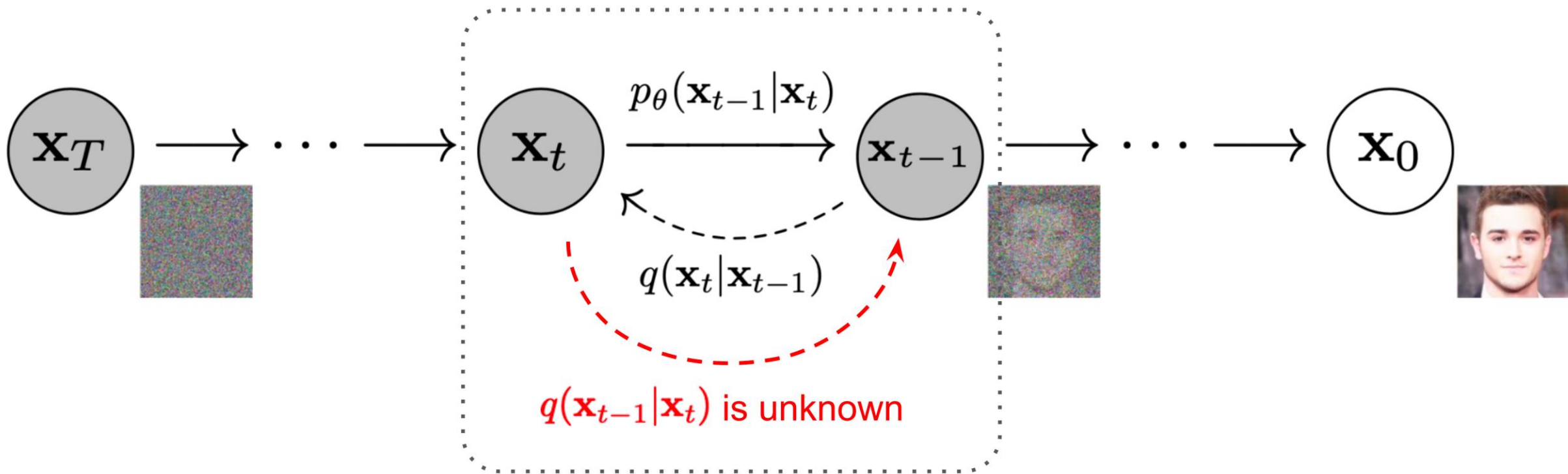
Tasks	Applications	Frame	Representative Methods
Molecule Modeling	Molecule Conformation Generation	SMLD	MDM [Huang <i>et al.</i> , 2022]
		DDPM	GeoDiff [Xu <i>et al.</i> , 2022], EDMs [Hoogeboom <i>et al.</i> , 2022], EEGSDE [Bao <i>et al.</i> , 2022], DiGress [Vignac <i>et al.</i> , 2022]
		SGM	Torsional Diffusion [Jing <i>et al.</i> , 2022], MOOD [Lee <i>et al.</i> , 2022], GDSS [Jo <i>et al.</i> , 2022], DGSM [Luo <i>et al.</i> , 2021a], DiffBridges [Wu <i>et al.</i> , 2022b]
	Molecular Docking	DDPM	FragDiff [Anonymous, 2023c], DiffLink [Igashov <i>et al.</i> , 2022], TargetDiff [Anonymous, 2023a], DiffBP [Lin <i>et al.</i> , 2022]
		SGM	DiffDock [Corso <i>et al.</i> , 2022]
Protein Modeling	Protein Generation	DDPM	SMCDiff [Trippe <i>et al.</i> , 2022], SiamDiff [Zhang <i>et al.</i> , 2022], DiffFold [Wu <i>et al.</i> , 2022a], ProSSDG [Anand and Achim, 2022], DiffAntigen [Luo <i>et al.</i> , 2022a], RFdiffusion [Watson <i>et al.</i> , 2022]
		SGM	ProteinSGM [Luo <i>et al.</i> , 2022a]
	Protein-ligand Complex Structure Prediction	DDPM	DiffEE [Nakata <i>et al.</i> , 2022]
		SGM	NeuralPLexer [Qiao <i>et al.</i> , 2022]

Fan, Wenqi, et al. *arXiv preprint* 2023.



■ Denoising Diffusion Probabilistic Models (DDPMs)

Use variational lower bound

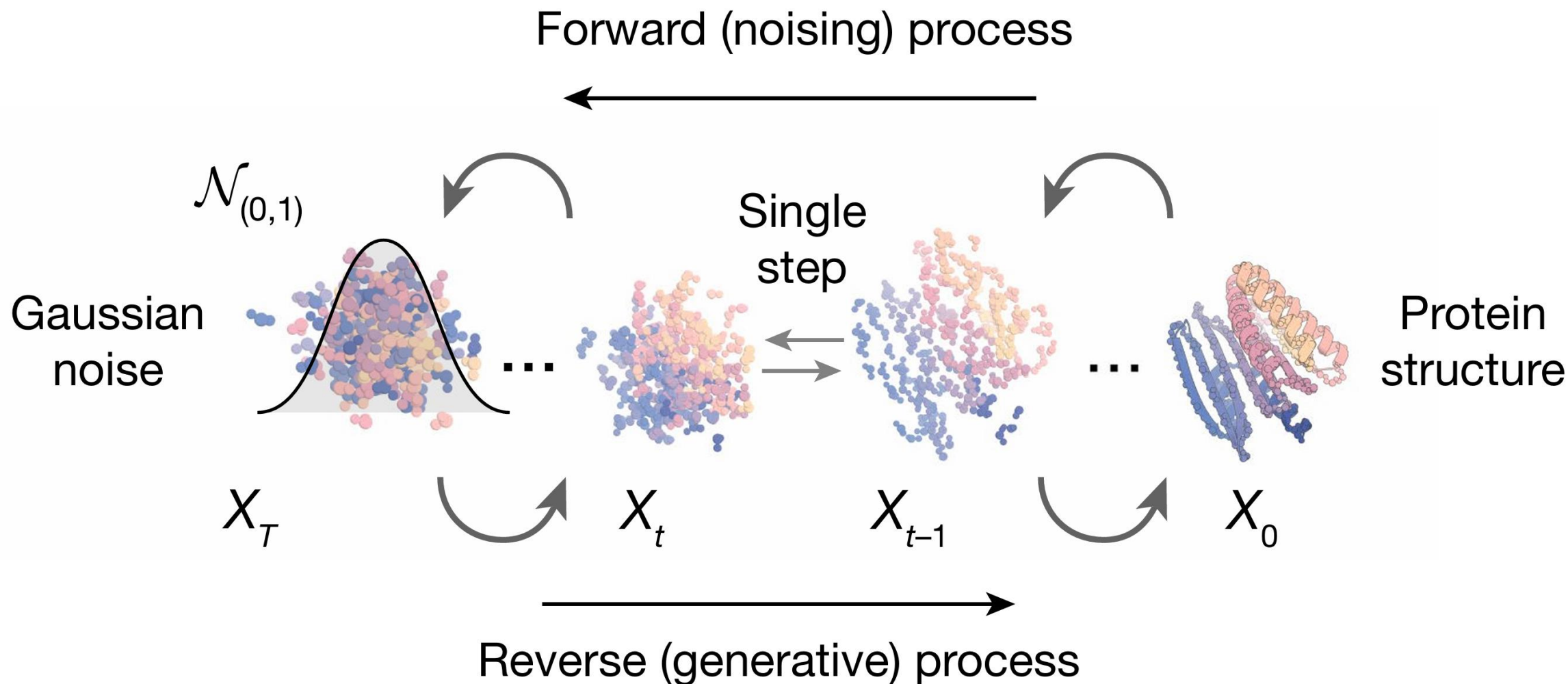


Ho, Jonathan, Ajay Jain, and Pieter Abbeel. NeurIPS 2020.

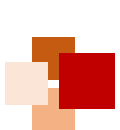
DDPMs for protein design



Diffusion model



Watson, J.L., Juergens, D., Bennett, N.R. et al. Nature 2023.



Article

De novo design of protein structure and function with RFdiffusion

<https://doi.org/10.1038/s41586-023-06415-8>

Received: 14 December 2022

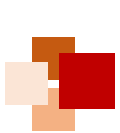
Accepted: 7 July 2023

Published online: 11 July 2023

Open access

Joseph L. Watson^{1,2,15}, David Juergens^{1,2,3,15}, Nathaniel R. Bennett^{1,2,3,15}, Brian L. Trippe^{2,4,5,15}, Jason Yim^{2,6,15}, Helen E. Eisenach^{1,2,15}, Woody Ahern^{1,2,7,15}, Andrew J. Borst^{1,2}, Robert J. Ragotte^{1,2}, Lukas F. Milles^{1,2}, Basile I. M. Wicky^{1,2}, Nikita Hanikel^{1,2}, Samuel J. Pellock^{1,2}, Alexis Courbet^{1,2,8}, William Sheffler^{1,2}, Jue Wang^{1,2}, Preetham Venkatesh^{1,2,9}, Isaac Sappington^{1,2,9}, Susana Vázquez Torres^{1,2,9}, Anna Lauko^{1,2,9}, Valentin De Bortoli⁸, Emile Mathieu¹⁰, Sergey Ovchinnikov^{11,12}, Regina Barzilay⁶, Tommi S. Jaakkola⁶, Frank DiMaio^{1,2}, Minkyung Baek¹³ & David Baker^{1,2,14}✉



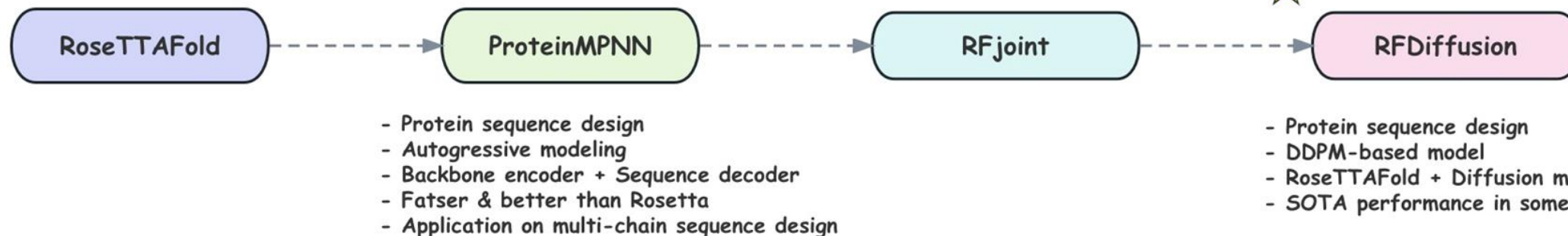


RoseTTAFold发展演变历程



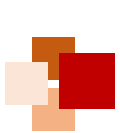
- Protein structure prediction
- Three-track NN (1D / 2D / 3D)
- SE(3)-equivariant transformer
- Structure predictions with accuracies approaching those of DeepMind in CASP14

- Protein sequence design
- Partial hallucination method
- Token masking task (like BERT)



- 在一众扩散 (Diffusion) 模型中，美国华盛顿大学 David Baker 团队的 RFdiffusion 十分耀眼
- 从 RoseTTAFold (RF) 的**从序列预测结构**，再到 ProteinMPNN 与 RFjoint 的**从结构预测序列**，再到 RFdiffusion，将以上两种功能有机结合在一起，最终实现了利用 diffusion 同时生成结构和序列的能力
 - 这个能力来自于**AF2/RF强大的 Language Model 框架**以及**Diffusion 强大的生成能力**

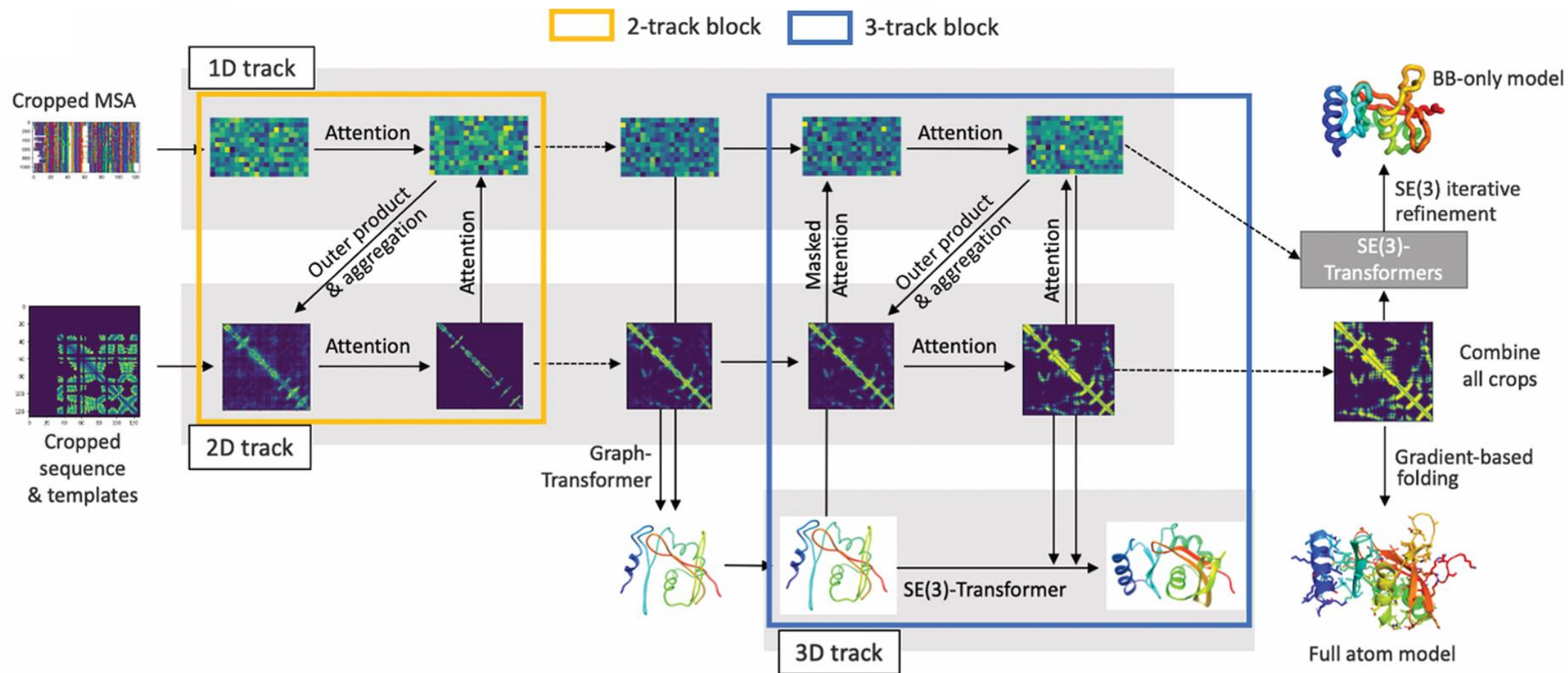




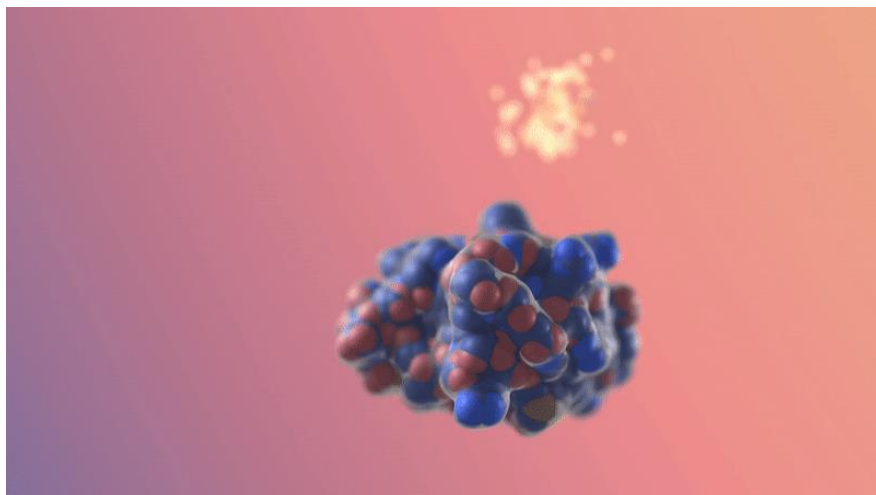
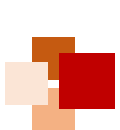
RoseTTAFold



上海科技大学
ShanghaiTech University



Baek et al., Science 373, 871–876 (2021)



图为基于给定蛋白结合域的
binder蛋白随机生成

团队排名	Group Name	Country	University or Company
1	Yang-Server	China	Shandong University, Nankai University, etc
	Yang		
2	UM-TBM	USA	University of Michigan, Michigan State University
3	PEZYFoldings	Japan	Infinite Curation Inc.
4	Dfolding	Korea	Hanyang University, Korea Institute for Advanced Study, etc
	Dfolding-server		
5	McGuffin	UK	University of Reading
6	MULTICOM	USA	University of Missouri
	MULTICOM_human		
	MULTICOM_refine		
	MULTICOM_egnn		
	MULTICOM_deep		
	MULTICOM_qa		
7	Kiharalab	USA, Japan	Purdue University, Tokyo Institute of Technology, etc
8	Manifold-E	China	DP technology
	Manifold		
9	BAKER	USA	University of Washington
10	ShanghaiTech	China	ShanghaiTech University

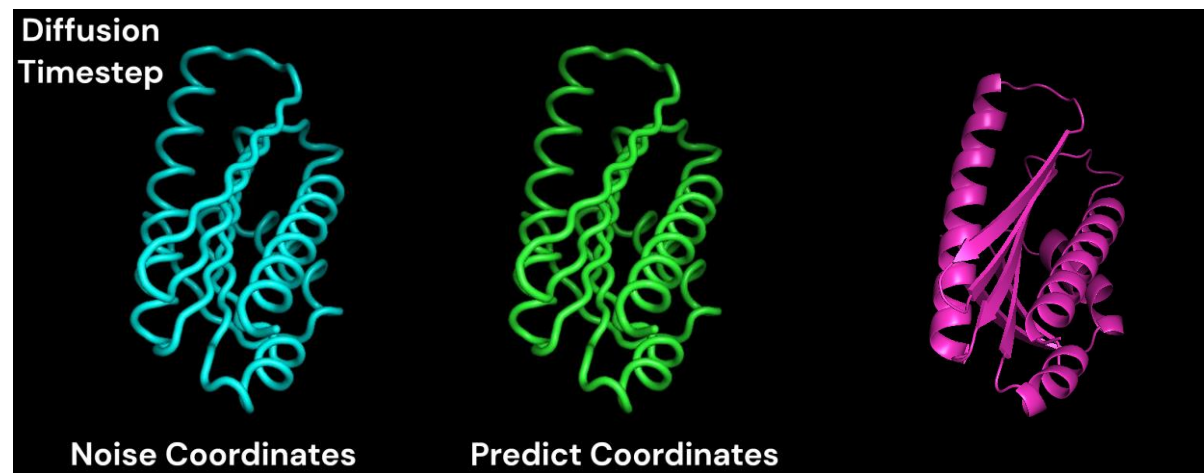
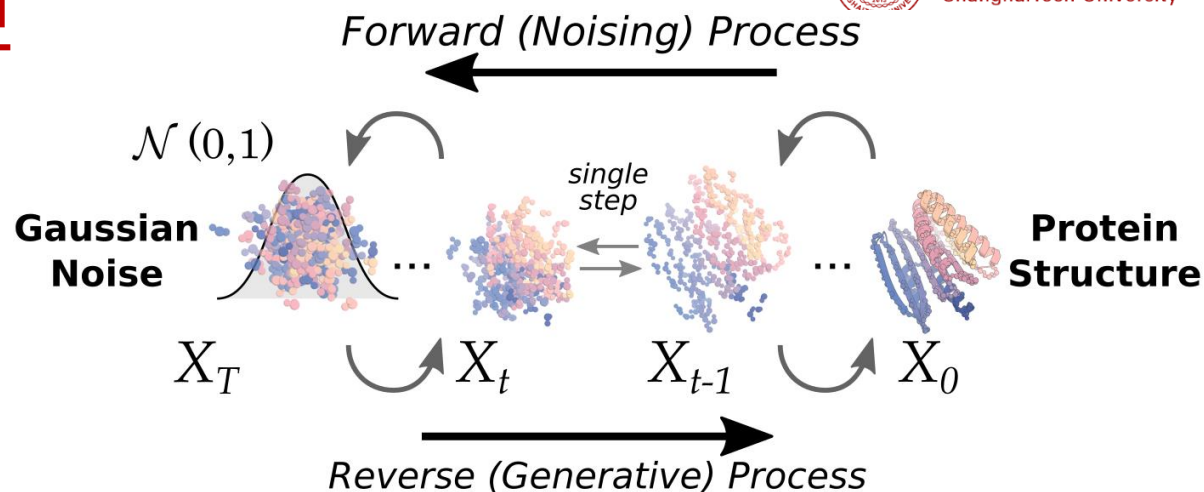
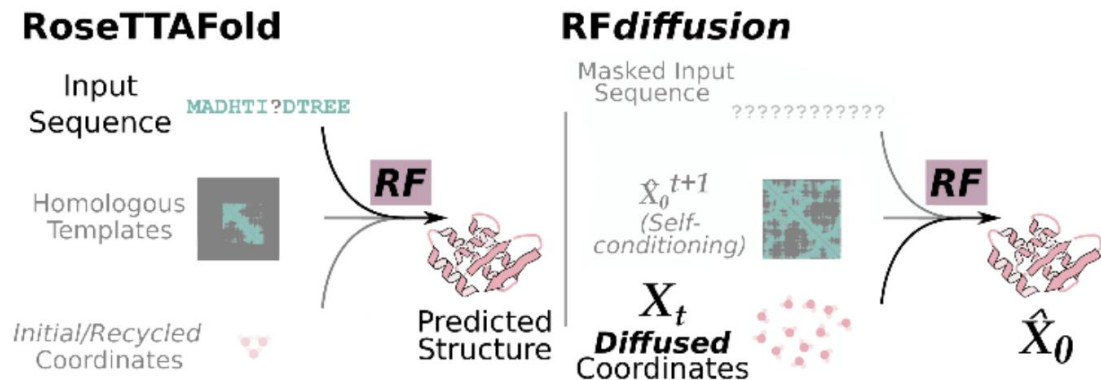
CASP15主赛道前10名

David Baker 团队的 RFdiffusion 模型在国际蛋白预测竞赛 CASP15的主赛道中取得全球第9名次。

一个生成模型能在预测竞赛中取得如此斐然成绩，令人印象深刻。



Main idea of RFdiffusion



1. RFdiffusion 将每一个扩散时间步称为一个 **Sampler** (共200步)
2. **Sampler** 由 **RoseTTAFold** 和一个 **denoising diffusion model** 构成
3. 在生成过程中，随着时间步降低，Denoiser 输出的**噪声坐标**会越来越接近**生成坐标**

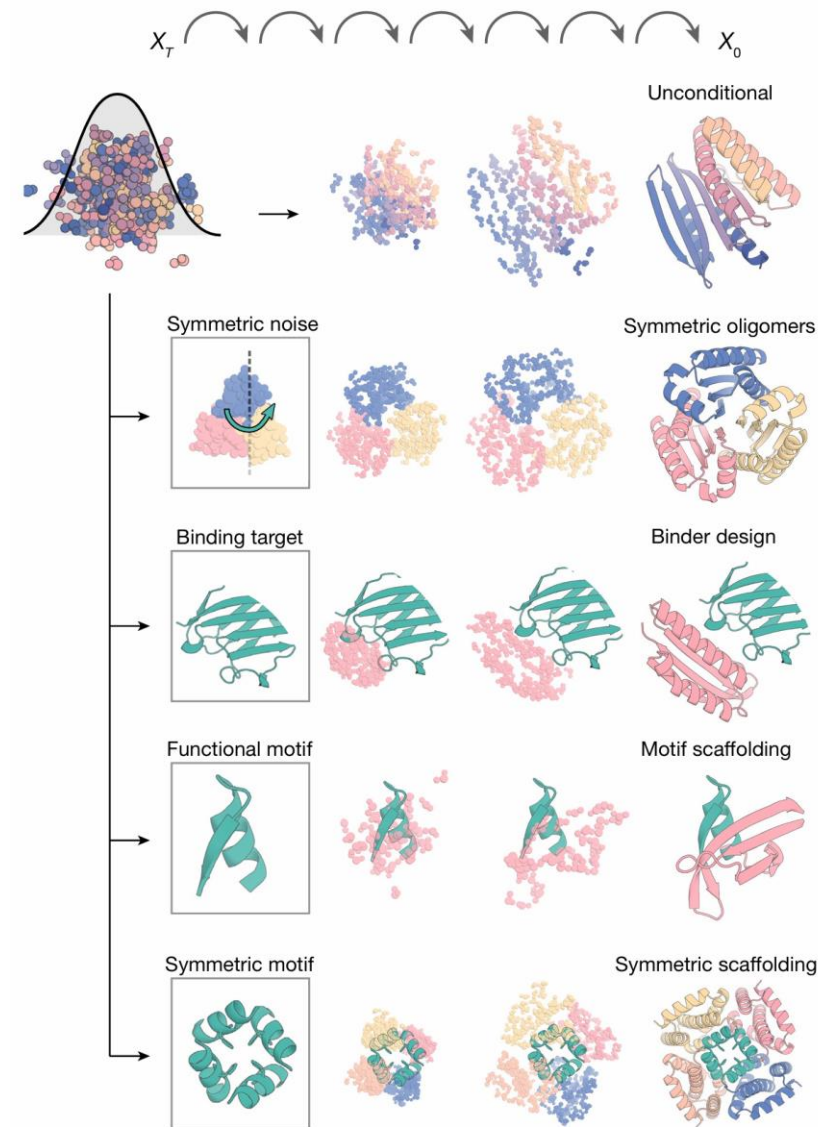
RFdiffusion Self-conditioning Forward演示。
反向则是随机结构生成的过程。



RFdiffusion



- **Key insight:** one can make the denoising process work much better by using a structure predictor as a component
 - RFDiffusion uses RoseTTAFold (RF) as a starting point, and tunes it for this application
 - Like AlphaFold 2, the recent version of RoseTTAFold (RF) uses “**recycling**”, i.e. run the prediction process several times, feeding the output coordinates as inputs to the next prediction. This inspired the use of RF in RFdiffusion
- The denoising (generative) process can use information of properties of the protein to be generated, known as “**conditioning**” to generate designs **with desired properties** (e.g. functional motifs, symmetry, binding target)



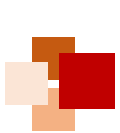


Why choose DDPMs for protein design?



- DDPMs have several properties well-suited to protein design
 - Highly diverse outputs
 - Conditioning generation at each step
 - Explicitly 3D structure modeling via **SE(3)-equivariant** DDPMs
- Previous DDPMs-related methods had limited success
 - Limited ability of the denoising networks to generate realistic protein backbones
 - No experimental validation
- Need **a powerful protein structure prediction framework** that can act as a denoiser

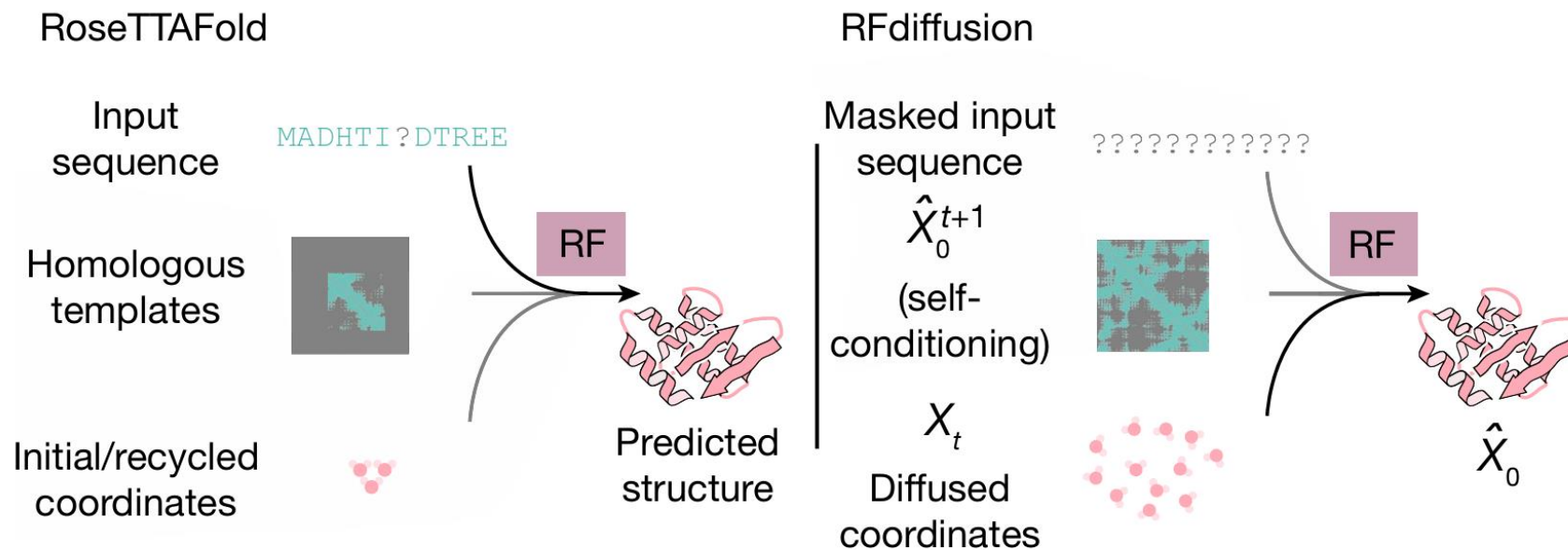




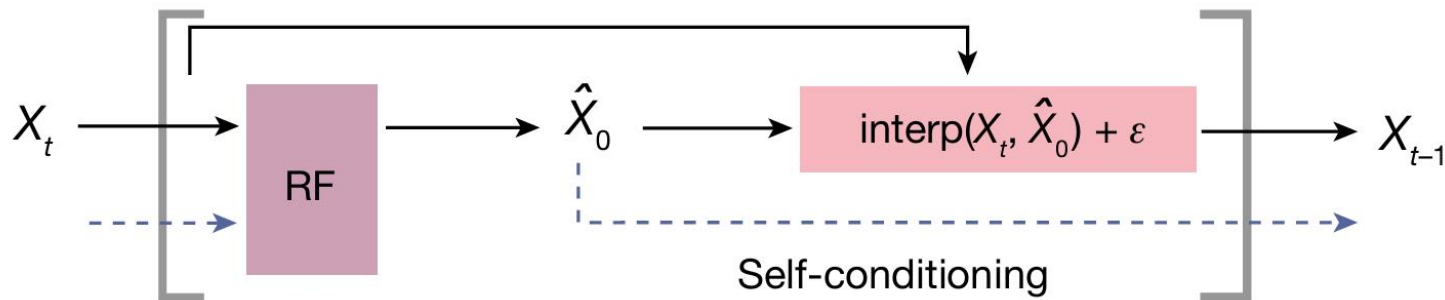
RoseTTAFold vs. RFDiffusion



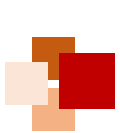
上海科技大学
ShanghaiTech University



Single RFDiffusion step



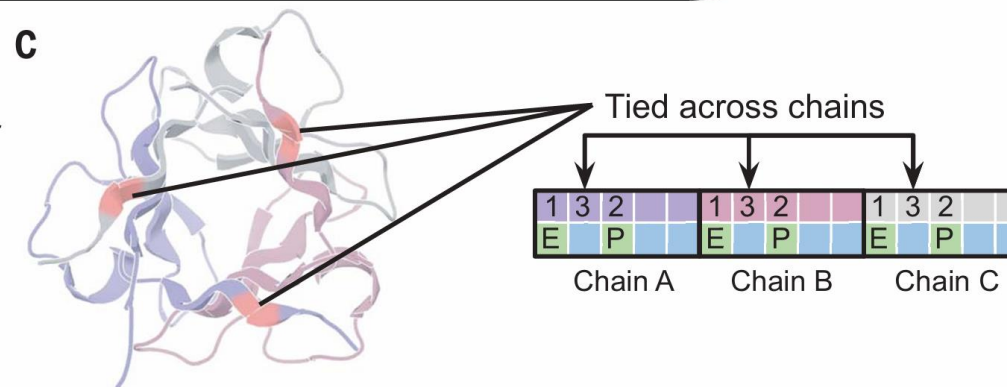
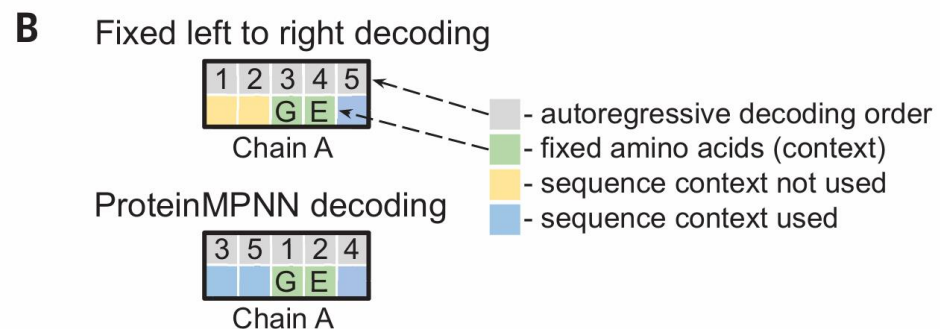
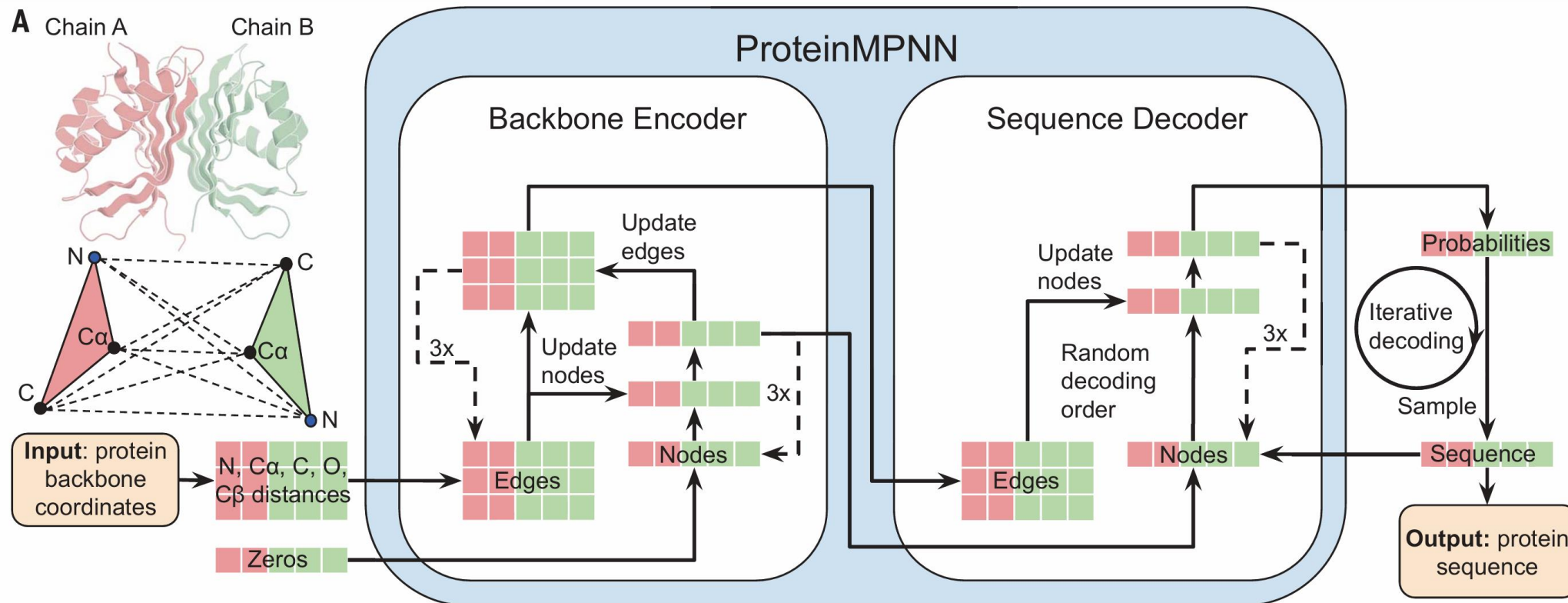
Watson, J.L., Juergens, D., Bennett, N.R. et al. Nature 2023.



ProteinMPNN



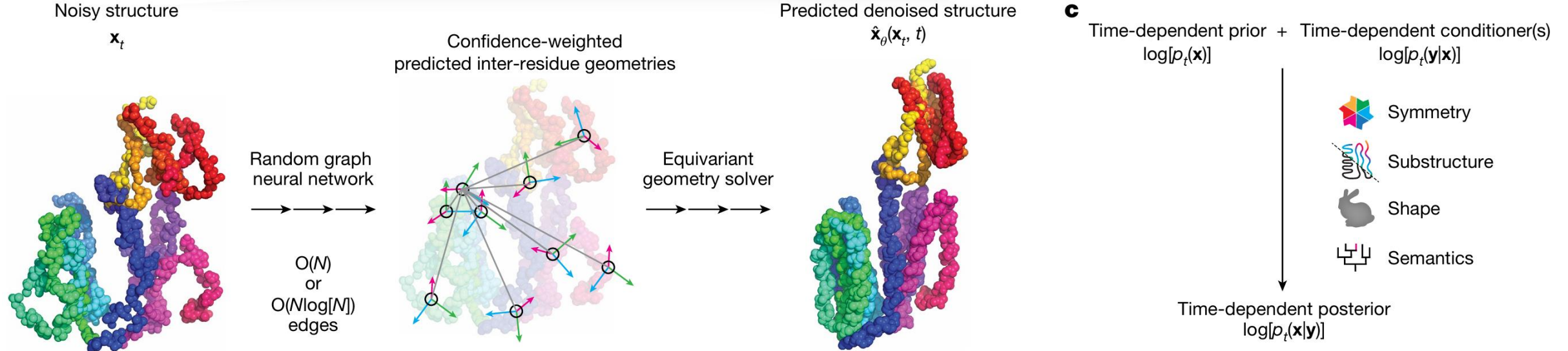
上海科技大学
ShanghaiTech University



Dauparas et al., Robust deep learning-based protein sequence design using ProteinMPNN, Science **378**, 49–56, 2022.



Other AI models for protein design



Ingraham, J.B., et al. Nature 623, 1070–1078, 2023 (**Chroma**)

- **Chroma**: a generative model for proteins and protein complexes that can directly sample novel protein structures and sequences. It can be conditioned to steer the generative process towards desired properties and functions, achieving protein design as **Bayesian inference under external constraints**.
- **ProteinDT**: a **multi-modal** framework that leverages **textual descriptions to guide protein design** through a three-step process: (1) aligning text and protein sequence representations, (2) generating protein representations from text, and (3) decoding these representations into protein sequences.



Recap and discussion



- In this lecture, we have learned:
 - Protein design can be considered an inverse process of protein structure prediction
 - There are in silico metrics and experimental approaches for evaluation of proteins generated by AI
 - Protein language models (based on pre-trained large language models such as BERT or GPT) can be used to design proteins
 - Diffusion models are underlying some most successful protein design AI models (e.g. RFdiffusion, Chroma)
 - Models for protein sequence design (e.g. ProteinMPNN) works with structure design models (e.g. RFdiffusion) in a pipeline
 - It is crucial, yet still challenging, to generate proteins of desirable properties (i.e. conditioning)
- Questions for discussion:
 - Can AlphaFold 3 be used for de novo protein design? Why or why not?
 - How to describe desired protein functions more precisely to more effectively guide the generative AI models?





References



- Madani et al. “Large language models generate functional protein sequences across diverse families” , Nature Biotechnology, 2022 (**ProGen**).
- Nijkamp E, et al. ProGen2: Exploring the boundaries of protein language models. Cell Systems, 14(11):968-978.e3, 2023 (**ProGen2**)
- Ho et al. “Denoising diffusion probabilistic models” , NeurIPS, 2020 (**DDPM**).
- J. L. Watson et al. “De novo design of protein structure and function with RFdiffusion” , Nature, Vol 620, pp. 1089-1100, 2023 (**RFdiffusion**).
- Dauparas J, et al.. Robust deep learning-based protein sequence design using ProteinMPNN. Science. 378(6615):49-56, 2022 (**ProteinMPNN**)
- Ingraham, J.B., et al. Illuminating protein space with a programmable generative model. Nature 623, 1070–1078, 2023 (**Chroma**)

